

DAY 06 NOTES

Excel Analysis – From Calculations to Business Conclusions

1. Day 06 Objective

The main objective of Day 06 was to develop an analyst mindset: calculate KPIs, compare performance, identify the strongest/weakest areas, drill down into the data, investigate possible drivers, and make a basic business conclusion.

2. Analytical Question Framework

A good analysis follows this flow:

Question → Calculation → Comparison → Finding → Business Conclusion

- Which country has the highest/lowest sales?
- Which quarter/month has the highest/lowest sales?
- Which branch performs best/worst?
- Which salesperson generates the highest sales?
- Which salesperson achieves the highest target percentage?
- Which product performs best/worst?
- Who/which area exceeded or missed target?
- Where is the largest positive or negative variance?

3. Core KPI Calculations

KPI	Formula	Meaning
Achievement %	$\text{Actual Sales} \div \text{Target} \times 100$	How much of the target was achieved
Variance	$\text{Actual Sales} - \text{Target}$	Absolute gap from target
Variance %	$(\text{Actual} - \text{Target}) \div \text{Target} \times 100$	Relative gap from target

Important relationship:

Achievement % = 100% + Variance %

Example: If Achievement = 85%, then Variance % = -15%.

4. KPI Example

Metric	Value
Actual Sales	₹850,000
Target	₹1,000,000
Achievement %	85%
Variance	-₹150,000
Variance %	-15%

Interpretation: The business achieved 85% of its target and is short by ₹150,000, which is a 15% negative variance.

5. Country Analysis

Country	Actual Sales	Target	Achievement	Variance	Variance %
Kenya	₹220K	₹200K	110%	+₹20K	+10%
Nigeria	₹180K	₹250K	72%	-₹70K	-28%
Tanzania	₹150K	₹150K	100%	₹0	0%
Uganda	₹100K	₹80K	125%	+₹20K	+25%

- Highest actual sales: Kenya
- Highest achievement: Uganda
- Largest positive absolute variance: Kenya and Uganda (tie at +₹20K)
- Biggest target-performance concern: Nigeria

Key lesson: 'Best' depends on the KPI. Actual Sales measures business volume, while Achievement % and Variance % measure performance relative to target.

6. Branch Analysis

Branch	Actual	Target	Achievement	Variance	Variance %
A	₹500K	₹450K	111.1%	+₹50K	+11.1%
B	₹300K	₹250K	120%	+₹50K	+20%
C	₹400K	₹500K	80%	-₹100K	-20%
D	₹200K	₹180K	111.1%	+₹20K	+11.1%

Branch B was selected as the strongest branch for target performance because it achieved 120% of target and had a +20% variance, even though Branch A had the same absolute +₹50K variance.

7. Product Analysis

Product	Actual	Target	Achievement	Variance	Variance %
X	₹600K	₹500K	120%	+₹100K	+20%
Y	₹400K	₹300K	133.3%	+₹100K	+33.3%
Z	₹250K	₹200K	125%	+₹50K	+25%

- Highest actual sales: Product X
- Strongest target performance: Product Y
- X and Y have the same absolute positive variance (+₹100K), but Y has stronger relative performance.

8. Salesperson Analysis

Salesperson	Actual	Target	Achievement	Variance	Variance %
Rahul	₹500K	₹400K	125%	+₹100K	+25%
Priya	₹450K	₹350K	128.6%	+₹100K	+28.6%
Amit	₹300K	₹200K	150%	+₹100K	+50%
Neha	₹600K	₹550K	109.1%	+₹50K	+9.1%

Neha generated the highest actual sales (₹600K), but Amit had the strongest target performance with 150% achievement and +50% variance %. Therefore, the answer depends on the KPI being evaluated.

9. Time-Based / Quarter Analysis

Quarter	Actual Sales	Target	Achievement	Variance	Variance %
Q1	₹800K	₹700K	114.3%	+₹100K	+14.3%
Q2	₹950K	₹1,000K	95%	-₹50K	-5%

Q3	₹1,100K	₹900K	122.2%	+₹200K	+22.2%
Q4	₹1,050K	₹1,000K	105%	+₹50K	+5%

- Q3 has the highest actual sales.
- Q3 has the highest achievement percentage.
- Q3 has the largest positive variance.
- Therefore, Q3 is the strongest overall quarter in this example.

10. Drill-Down Analysis

Do not stop at a high-level finding. If Q3 is the best quarter, investigate why.

Recommended drill-down:

Q3 → Country → Branch → Product → Salesperson

- Compare Q3 branches to identify which branch contributed most.
- Check salesperson performance to find key contributors.
- Check product-level sales to identify the products driving growth.
- Compare countries/regions to identify geographic drivers.
- Compare Q1–Q4 trends and, if available, compare with the previous year.

11. Finding vs Cause

Type	Example
Finding	Q3 has the highest sales.
Possible cause	A particular branch or product drove Q3 growth.
Verified cause	Data confirms that the branch/product contributed most of the increase.

Important: Do not assume causation just because two events happen at the same time. Use data to verify the driver.

12. Investigating a 20% Increase in Q3

If Q3 sales increased by 20%, the first step is to investigate whether the growth is:

- Across all quarters or only Q3.
- Across all branches or concentrated in a few branches.
- Across all products or driven by specific products.
- Across all salespeople or driven by specific individuals.
- Related to promotions, discounts, pricing, festivals/holidays, seasonality, or new product launches.
- Related to a major customer/order or a market/competitor change.

13. Day 06 Analysis Framework

Use this framework in future Excel analysis:

1. Calculate – Create the required KPI.
2. Compare – Compare countries, branches, products, people, or time periods.

3. Identify – Find the highest, lowest, positive, negative, or unusual result.
4. Drill Down – Break the result into smaller dimensions.
5. Investigate – Look for the actual business driver.
6. Conclude – State the finding clearly and avoid unsupported assumptions.

14. Key Learning Points

- Highest sales does not automatically mean best performance.
- Achievement % tells you how well actual sales performed against target.
- Variance tells you the absolute target gap.
- Variance % tells you the relative target gap.
- Two people/branches can have the same variance but different variance percentages.
- A strong analyst asks 'Why?' after identifying 'What happened?'
- High-level analysis should lead to drill-down analysis.
- External and seasonal factors should be considered, but they must be supported by evidence.
- A business conclusion should be based on the relevant KPI and business objective.

15. Day 06 Final Takeaway

The main skill developed on Day 06 was moving from simple Excel calculations to analytical thinking. The goal is not only to calculate numbers, but to understand what they mean, compare performance, identify important patterns, investigate the drivers, and communicate a clear business conclusion.